

CHAPTER 48

Clinical trials for tinnitus: study populations, designs, measurement variables, and data analysis

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Abstract: We review a few issues related to clinical trials for treating patients with tinnitus, including the study population, design, choice of measurement variables, and some new approaches to data analysis. We emphasize the importance of being aware of different subgroups of tinnitus patients, and that patients who have had tinnitus for less than 6 months could be more amenable to treatment than patients who have had their tinnitus for a longer period. We distinguish the tinnitus itself, from the reactions to the tinnitus. When the treatment is intended to reduce the tinnitus, we recommend measuring the magnitude of the tinnitus. We provide arguments and data to support the use of the Tinnitus Handicap Questionnaire as a measure of the reaction to the tinnitus. We suggest that the current quality of life measures are not valid for measuring lifestyle effects of alleviating tinnitus. Because tinnitus likely has different subgroups, and because tinnitus affects people differently, we believe data analysis should emphasize individuals, not groups. A clinically meaningful effect should represent a valid and reliable statistical change for an individual.

Keywords: tinnitus; clinically significant; tinnitus handicap questionnaire; tinnitus severity; tinnitus measurement

Study population

Types of tinnitus

Tinnitus can be conductive or sensorineural, analogous to hearing loss (Tyler and Babin, 1993). Conductive (others prefer objective) tinnitus is transmitted to the cochlea by conduction in body tissue, and is typically of vascular or muscular cause whereas sensorineural (others prefer

subjective) tinnitus is not associated with a physical sound but caused by abnormal neural activity generated in the ear, the auditory nerve, or the central nervous system (see Chapter 1).

In the following sections we will focus on sensorineural tinnitus (although our comments can be applied to both forms). It is generally agreed that there are several kinds of sensorineural tinnitus but presently it is not clear how to distinguish between their different forms. Stouffer and Tyler (1990) estimated that approximately 1/3 of 528 tinnitus patients reported that their tinnitus was not present the entire day, and more than 20%

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reported it changed in pitch or loudness. Intermittent tinnitus can be even more annoying than continuous tinnitus if it is unpredictable. If tinnitus is intermittent or changing, it is necessary to quantify the magnitude of the tinnitus with multiple baseline measures and to consider the severity of tinnitus (before and after treatment) estimated over several days or weeks.

Exclusion criteria

Patients who cannot provide valid and reliable data should be excluded from studies. Obviously, it is important to ascertain as clearly as possible that any changes can be attributable to the treatment under study, and not due to some other variable. Therefore, patients undergoing other tinnitus treatments, such as counseling and sound therapies, medications, and dietary supplements should be excluded or required to stop treatment for a sufficient time. Similarly, patients facing other major health or lifestyle challenges, such as a major surgery, psychological counseling for depression, or divorce, should also be excluded.

Subgroups

Tinnitus is a symptom that may have many different causes and is likely coded in the nervous system by many different mechanisms. Therefore, any particular treatment may only be effective on a particular subgroup of individuals (Tyler, 1992).

Distinguishing between different subgroups of individuals with tinnitus is not self-evident, but likely include cause (e.g. noise, head injury, drug induced, aging), psychoacoustical behavior (e.g. post-masking categories or tonal masking patterns), and perceptions (tone-like vs. noise-like).

One subgroup might be patients who can modulate their tinnitus with muscular, proprioceptive, or perhaps visual stimulation (Cacace et al., 1999; Sanchez et al., 2002; Levine et al., 2003). We believe it is important to distinguish those modulations, which could be mediated by middle-ear muscle contraction (and therefore have their influence through the cochlea) from those maneuvers that are initiated by non-auditory neural

pathways at the cochlear nucleus and above (e.g. Møller et al., 1992; Kaltenbach et al., 2002; Shore, 2004).

Duration of tinnitus

There are two opposing matters in considering the duration of tinnitus as an inclusion criterion; (1) it is likely that the neural representation of tinnitus can more likely be altered if the treatment is applied as soon as possible after the neural code for the tinnitus is formed (see Chapter 2); and (2) there may be spontaneous recovery among some patients during the first few months. Additionally, even after the magnitude of the tinnitus becomes constant, the annoyance of the tinnitus can decrease over several months; an outcome sometimes termed “habituation”. The first consideration above suggests treating the patient as soon as possible. The second suggests excluding patients who have had tinnitus for less than about 1 year. We suggest treating the patient as soon as possible since it cannot be predicted which patients would experience spontaneous recovery, and by waiting an opportunity for successful treatment may be missed (see also Chapter 22). In a crossover design, and in a parallel group design, concerns about spontaneous recovery can be partially overcome by ensuring that the duration of tinnitus is matched across groups. We suspect that the number of patients with spontaneous recovery is small, and this consideration is offset by the potential to uncover a major effect of a treatment in patients with newly acquired tinnitus.

Crossover or parallel groups

When conducting tinnitus trials, crossover designs and parallel groups designs are often considered. In a crossover design, participants are recruited to receive multiple treatments. They will first receive one treatment where results are recorded. A wash-out period is then required to allow the first treatment to leave the subject’s system. After the washout, the subject receives the second treatment where results are recorded. The term ‘parallel groups’ represents two independent groups who

each receive separate treatments. There is no need for a washout period in this case.

A crossover design has many advantages over parallel groups. One of the biggest advantages is the ability to control for subject level variables. When a subject is able to serve at her or his own control, many extraneous variables that may have an impact on the group influence are accounted for, allowing the analysis to focus on the group effect. A crossover design also requires a smaller sample size for similar power. Only one group of participants needs to be recruited instead of two independent groups.

There are potential problems associated with a crossover design as well. First, the treatment might eliminate the tinnitus, so crossover is not possible. Second, the washout period might be unknown or very long. A very long washout period delays the study, and might increase dropouts or allow time-course natural changes to the tinnitus to arise.

It is important in parallel group designs to have an equal number of participants per group. This allows for one-to-one matching of the participants. Careful consideration is needed on the best approach to matching since it is not known how similar the matching must be and what are the important variables. Indeed, some important variables may not be known. For tinnitus, it is likely important to match on cause and duration of tinnitus, and possibly level of hearing loss. Of course, the degree of matching should be reported in publications.

Choice of measurement variables

A decision needs to be made whether the treatment is aimed at decreasing the tinnitus or the effects of the tinnitus. Figure 1 shows an adaptation of the model proposed by Dauman and Tyler (1992) to help conceptualize factors related to measuring and understanding tinnitus and its effects. The perception of the tinnitus (what it sounds like) leads to a reaction, but this reaction is influenced by an individual's psychology. Measures of the tinnitus itself are less influenced by these psychological factors.

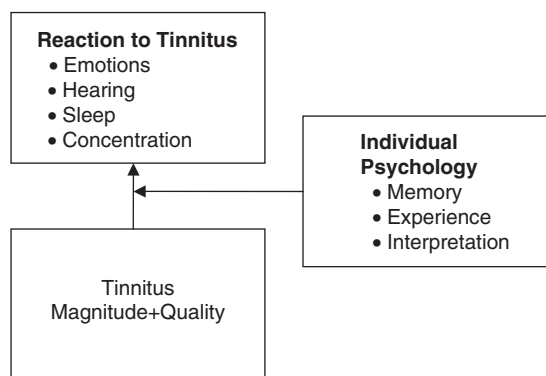


Fig. 1. Flowchart showing how the characteristics of tinnitus (its magnitude and quality) influence the reaction to the tinnitus, by are impacted by individual factors.

For an individual in a clinical trial, it should be expected that if the loudness of the tinnitus is decreased, it will be less annoying. Some have argued that the qualities (such as the loudness) of the tinnitus are unrelated to the annoyance. We believe this is not necessarily true. As Dauman and Tyler suggested, this relationship is influenced by the patient's psychology. Therefore, when examining the relationship across individuals the correlation is weak. But even then, patients who *report* that their tinnitus is louder tend to report their tinnitus as more annoying (see Fig. 2).

Any measurement must be valid (measure what it purports to measure), reliable, and sensitive (able to document changes). Objective measures, not requiring a judgment from the patient or experimenter, are not yet available for tinnitus. When objective measures are available, caution will be necessary as it is often difficult to interpret objective measures (their validity is unclear).

Primary measure

A single primary measurement is needed to determine if the treatment was effective, and if so, how effective. This should be decided before the clinical trial. Treatments are usually aimed at reducing the magnitude of the tinnitus (e.g. a drug treatment or electrical stimulation), or at reducing a patient's reaction to the tinnitus (e.g. a counseling

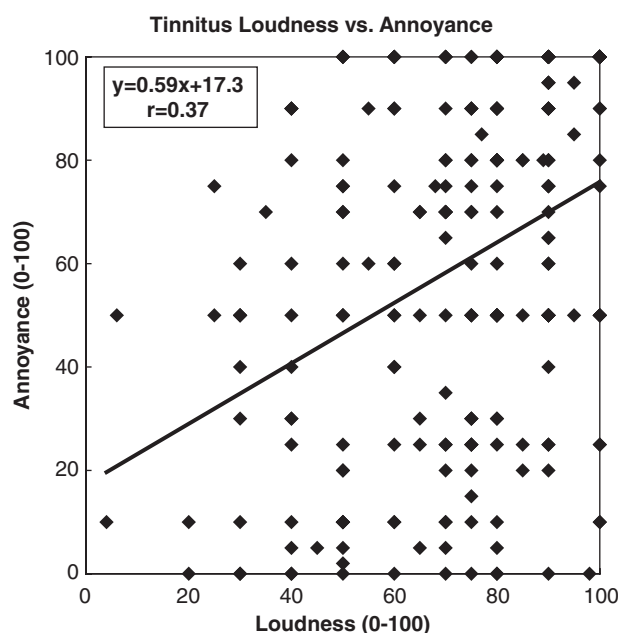


Fig. 2. The relationship between tinnitus loudness versus annoyance. Across different individuals, patients who rank their tinnitus as louder tend also to rank their tinnitus as more annoying. Individual difference in interpreting the scales, and individual psychological factors (see Fig. 1) influence this relationship when compared across individuals.

procedure). Measuring the reactions to the tinnitus are more likely to be influenced by non-treatment factors, and are less desirable when treatments are aimed at reducing the magnitude of the tinnitus.

Secondary measures

Secondary measures are not used to test the primary hypothesis. There might be a secondary hypothesis, for example that sleep problems are, or are not, affected by the treatment. Other common secondary measures might include depression, anxiety, or quality of life (see below). If the primary aim is to reduce the magnitude of the tinnitus, it would also be of interest to know if the reaction to the tinnitus is also reduced.

Predictor variables

In part because there are subgroups of tinnitus patients, it would be very useful to know which subgroups were most likely to benefit from

treatment. One approach is to select a large number of variables likely to separate tinnitus into groups, such as by etiology (e.g. noise, aging, head injury, drugs), biographical facts (e.g. age, gender), tinnitus history (e.g. ear, severity, duration), tinnitus quality (e.g. tonal, noise-like, rhythmic), and by the severity of depression and anxiety. [Stouffer and Tyler \(1990\)](#) provide such a questionnaire, as well as data from over 500 patients attending an otology/audiology clinic. Other variables might include the results of brain imaging and genetics.

Measures of the magnitude of the tinnitus

Options for measuring tinnitus magnitude include estimates of the minimum noise level required to mask the tinnitus or estimates of loudness. Masked thresholds, even for tinnitus, tend to have high test-retest reliability ([Small and Tyler, 1978](#)). This is a potential advantage over loudness measurements. However, the tinnitus of some patients

Table 1. Suggested single questions to evaluate loudness, annoyance, and the percentage of time tinnitus is present

1. Loudness

I want you to rate the loudness of your tinnitus during the last 2 weeks on a scale from 0 to 100; 0 representing the softest tinnitus you can imagine and 100 representing the loudest tinnitus you can imagine. Intermediate numbers represent intermediate loudness ratings.

2. Annoyance

I want you to rate the annoyance of your tinnitus during the last 2 weeks on a scale from 0 to 100; 0 representing the least annoying tinnitus you can imagine and 100 representing the most annoying tinnitus you can imagine. Intermediate numbers represent intermediate annoyance ratings.

3. Percentage of time present

During the time you are awake, what percentage of time is your tinnitus present?

Assign a value from 0 to 100%.

Source: Adapted from Stouffer and Tyler (1990).

cannot be masked at all (e.g. Tyler et al., 1984) and this would result in missing data. Two options for quantifying the loudness of tinnitus include using responses to questions (e.g. ratings from 0 to 100 of loudness or an equivalent visual analog scale) (see Table 1), or results of loudness balancing between the tinnitus and a matching signal (perhaps converting to sones) (e.g. Tyler and Conrad-Armes, 1983; Henry et al., 1999). Changes in hearing threshold might influence the interpretation of these test results, and it is therefore necessary to measure the thresholds to the test stimuli.

Measures of the reaction to tinnitus

Assessing patients' reactions to their tinnitus could be approached by either single questions or questionnaires.

For single questions, we suggest a focus on *loudness* and *annoyance* (consistent with Fig. 1) (Table 1). The term severity could be applied to either loudness or annoyance, but would be difficult to interpret. Stouffer and Tyler (1990) also suggested two other measures; the percentage of time tinnitus was present during the time awake, and the days per month that tinnitus is bothersome.

Using a scale from 0 to 100 is similar to using a visual analog scale, but has the practical advantage that it is not necessary to measure the distance

along a line. Patients are also familiar with the decimal system and this is an advantage over some questionnaires, where patients may have difficulty interpreting labeled scoring options. Handscomb (2006) reported that more than 1/3 of patients either filled in the Tinnitus Handicap Inventory (Newman et al., 1996) questionnaire incorrectly or not at all!

Another useful approach is to use individual questions from established questionnaires. This approach is illustrated in Fig. 3, which also compares the advantages in sensitivity of a 0–100 interval scale compared to a 3-point ordinal scale. We have selected individual patient data, combined from several studies in progress, showing pre-treatment and 6-month post-treatment scores on some individual questions of the Tinnitus Handicap Questionnaire (Kuk et al., 1990) and Tinnitus Handicap Inventory (Newman et al., 1998). We have selected single questions on concentration (Figs. 3a, b), hearing (Figs. 3c, d), and sleep (Figs. 3e, f). To the right of each panel, individual results are shown for patients who have multiple measures over time. In each pair of panels, the larger scale and better sensitivity of the Tinnitus Handicap Questionnaire allows the observation of smaller changes. Whenever an individual question is used, it is necessary to establish test–retest reliability.

Several questionnaires have been developed for tinnitus (see reviews by Tyler, 1993 and Noble, 1998). We believe the Tinnitus Handicap Questionnaire (Kuk et al., 1990) is perhaps the most widely used (e.g. Dobie et al., 1992; Dauman et al., 1993; Newman et al., 1994, 1995; Meric et al., 1997, 1998; Henry and Wilson, 1998; Higgins et al., 2002; Mrena et al., 2002; Bouscau-Faure et al., 2003; Henry et al., 2003, 2006; Robinson et al., 2003, 2005; Londero et al., 2006) and has been translated into other languages (e.g. Meric et al., 1997; Bouscau-Faure et al., 2003). This questionnaire was developed in two stages. First 87 questions were administered to 100 patients. Second, based on sensitivity and independence, a subset of questions was then tested on 319 patients. Also unlike other questionnaires, the test–retest reliability was evaluated by independent studies (e.g. Newman et al., 1995).

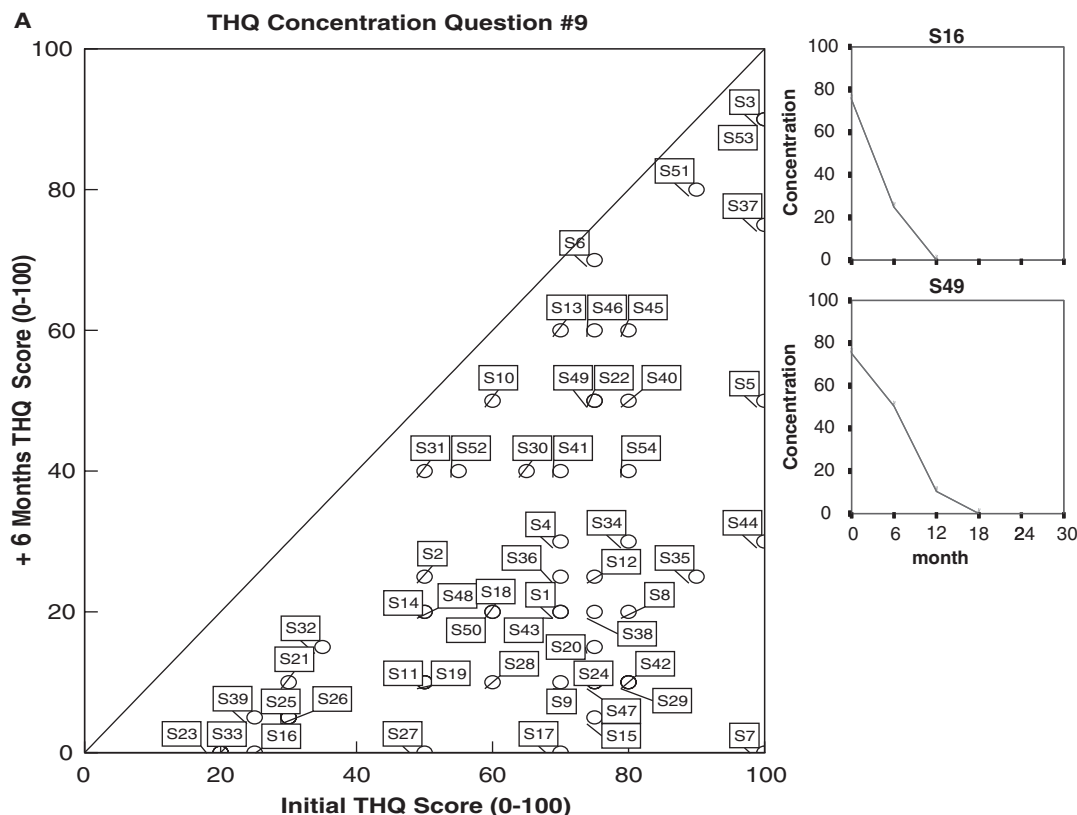


Fig. 3. Responses to individual questions on the THQ and THI before and after counseling and sound therapy. In each panel, the left scattergram shows scores pre and post 6-months treatment. The right figures show performance over time for two individuals. (A) THQ concentration question no. 9. (B) THI concentration question no. 1. (Yes to Yes: S3 S4 S6 S13 S37 S40 S51; Sometimes to Sometimes: S2 S8 S9 S10 S14 S27 S29 S32 S39 S41 S42 S43 S48 S50 S52; No to No: S11 S26; Yes to No: S7 S17; Yes to Sometimes: S1 S5 S19 S20 S22 S28 S30 S31 S34 S35 S36 S38 S44 S46 S47 S49 S53 S54; Sometimes to No: S12 S15 S16 S18 S21 S23 S24 S25 S33 S45). (C) THQ hearing question no. 4. (D) THI hearing question no. 2. (Yes to Yes: S1 S3 S9 S12 S14 S27 S36; Sometimes to Sometimes: S2 S7 S13 S17 S19 S24 S31 S34. No to No: S25; Yes to No: S15 S16 S22 S30 S40; Yes to Sometimes: S4 S5 S6 S11 S18 S20 S23 S33 S35 S37 S39 S41. No to Sometimes: S21; Sometimes to No: S8 S10 S26 S28 S29 S32 S38). (E) THQ sleep question no. 16. (F) THI sleep question no. 7.

Quality of life questionnaires

An effective treatment should ultimately impact on the patients' quality of life. However, we believe that the primary measure should be the tinnitus or its effects because there are other variables affecting the patients' emotions and social life (true for all of us). For example, the tinnitus magnitude and annoyance might have decreased, but because the person was involved in a car accident or got in a fight with friends that week, the impact of the effectiveness of the treatment was not observed. The influence of other confounding variables is

greatest the furthest away one moves from the tinnitus magnitude.

A second concern about these quality of life measures is their validity. How does one know which questions are relevant to the quality of life? In our opinion, several of the questions in many of these scales focus on mobility and self care. How might their validity be understood in the context of tinnitus? These are difficult questions. If we use quality of life measures that do not have appropriate validity, we could make serious mistakes in the interpretation of the results of tests using such questionnaires. We suggest not using quality of life

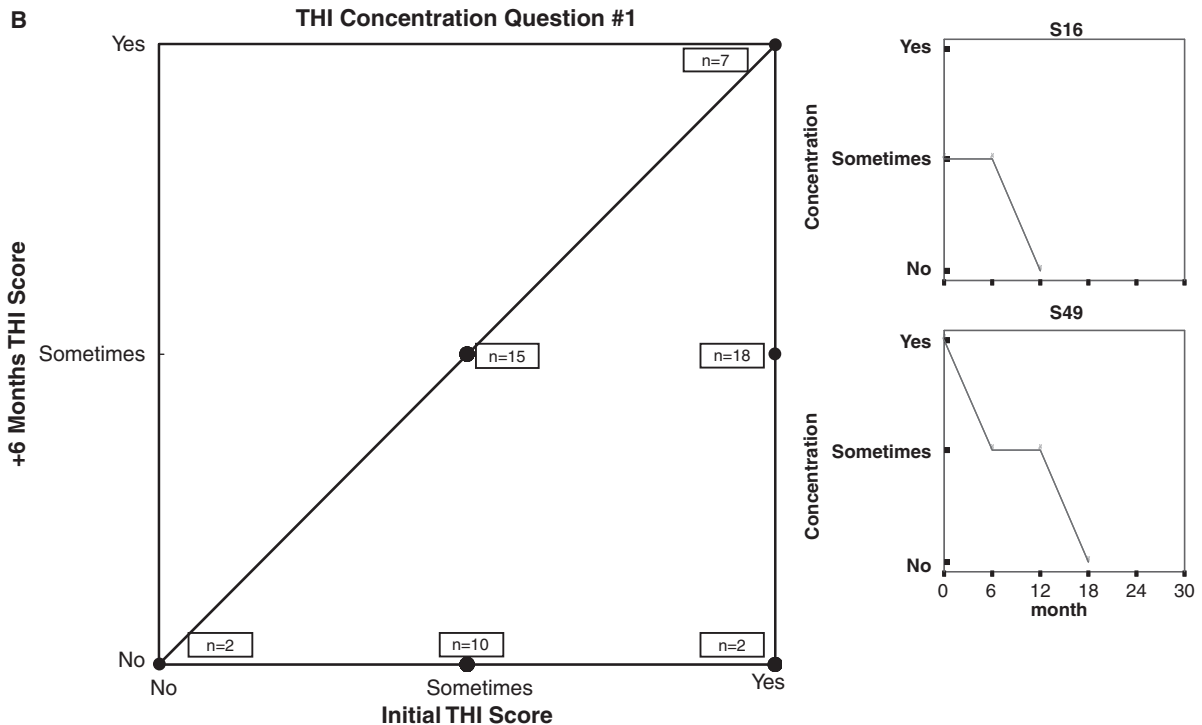


Fig. 3 (Continued)

measures until convincing data can be presented regarding validity.

Data analysis

Group versus individual data

There is a long-standard accepted approach to hypothesis testing that compares averages of groups. The assumption is that differences across participants are due to individual variation and measurement error. In an ideal world everyone would react the same way to the given treatment. We know in practice with tinnitus patients that individuals do not react the same as their prior data tells us they should. With the large degree of individual variation, group level statistical significance is difficult to attain unless there are large data sets. The standard hypothesis testing based on groups provides guidance regarding the power, effect size to be expected, and number of participants needed within a group to determine differences between

groups. There are instances, however, when individual data is much more informative.

It might be considered blasphemy to suggest that data analysis based on group statistics might not always be appropriate. The tinnitus of different patients may very well be different. The causes and mechanisms of the tinnitus differ and patients' reactions are different. We believe that it is more appropriate to examine individual results than group results. As an alternative, one could focus on the number of patients who benefited from a procedure. If important, one could specify the number of patients who were helped by a certain amount. In this context it is important to ascertain what constitutes a significant improvement for an individual.

Clinically relevant effect

People often attempt to distinguish a clinically relevant effect from a statistically significant one.

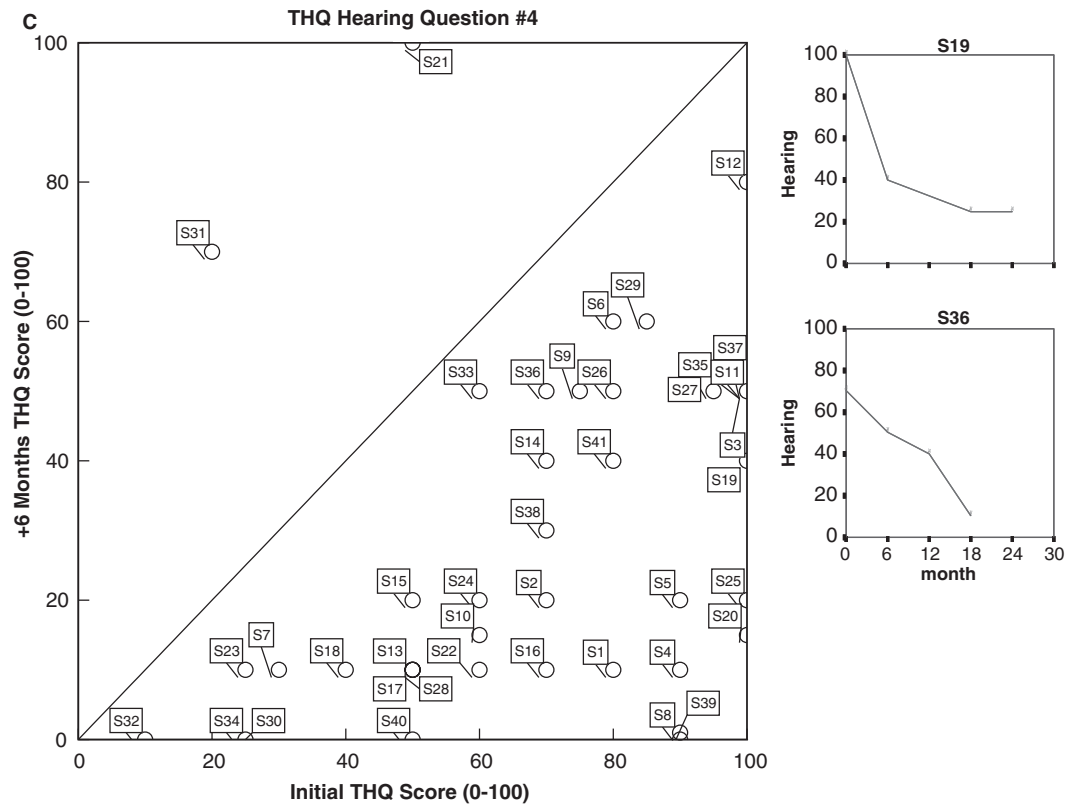


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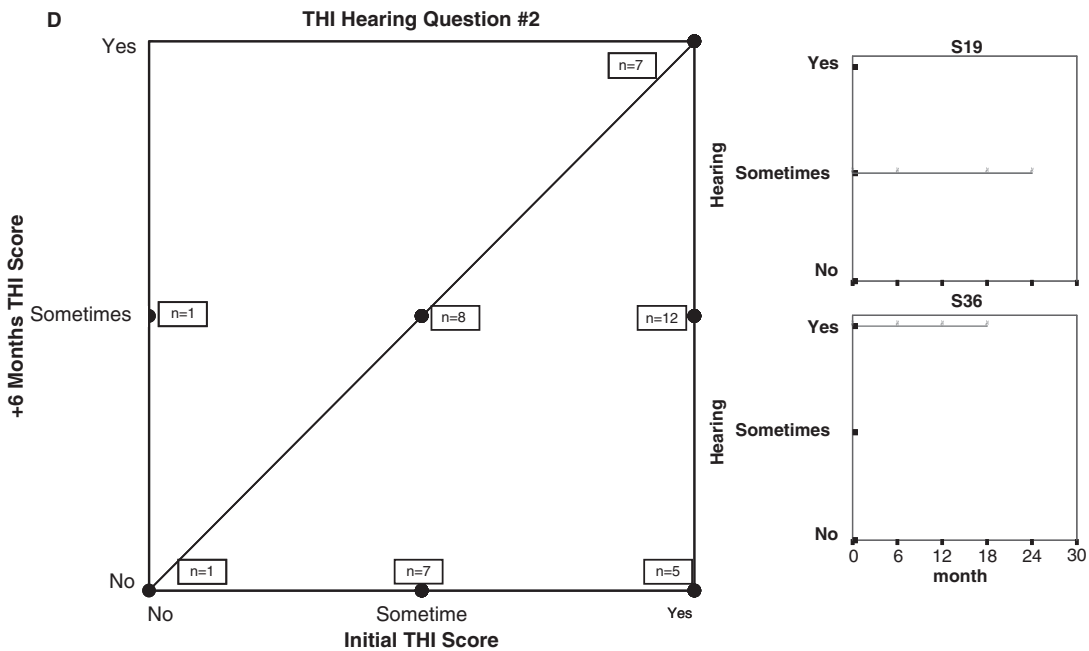


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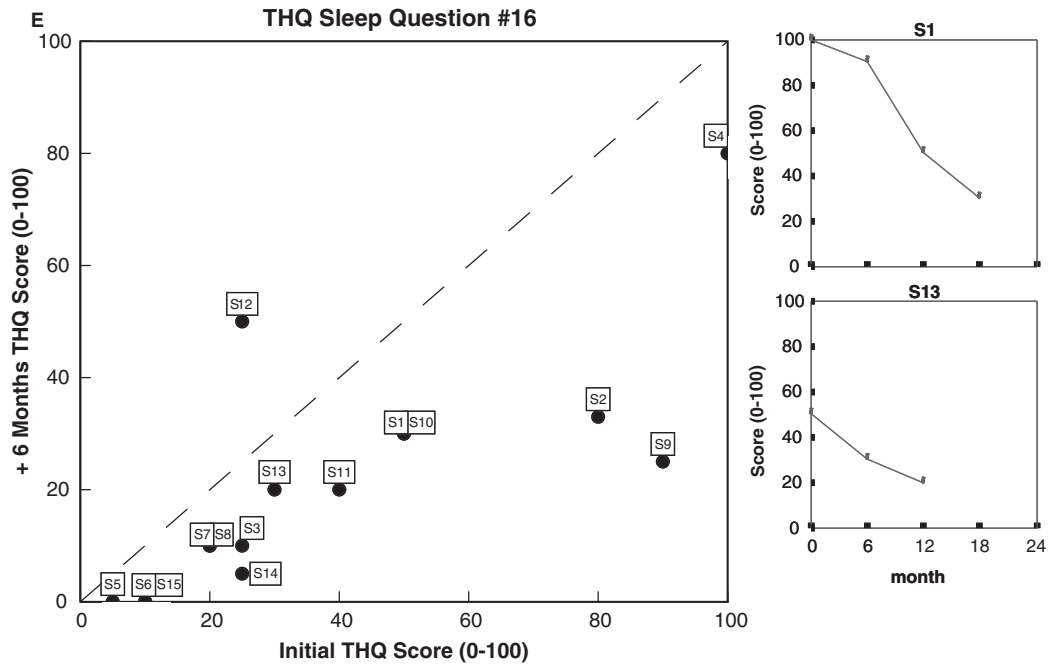


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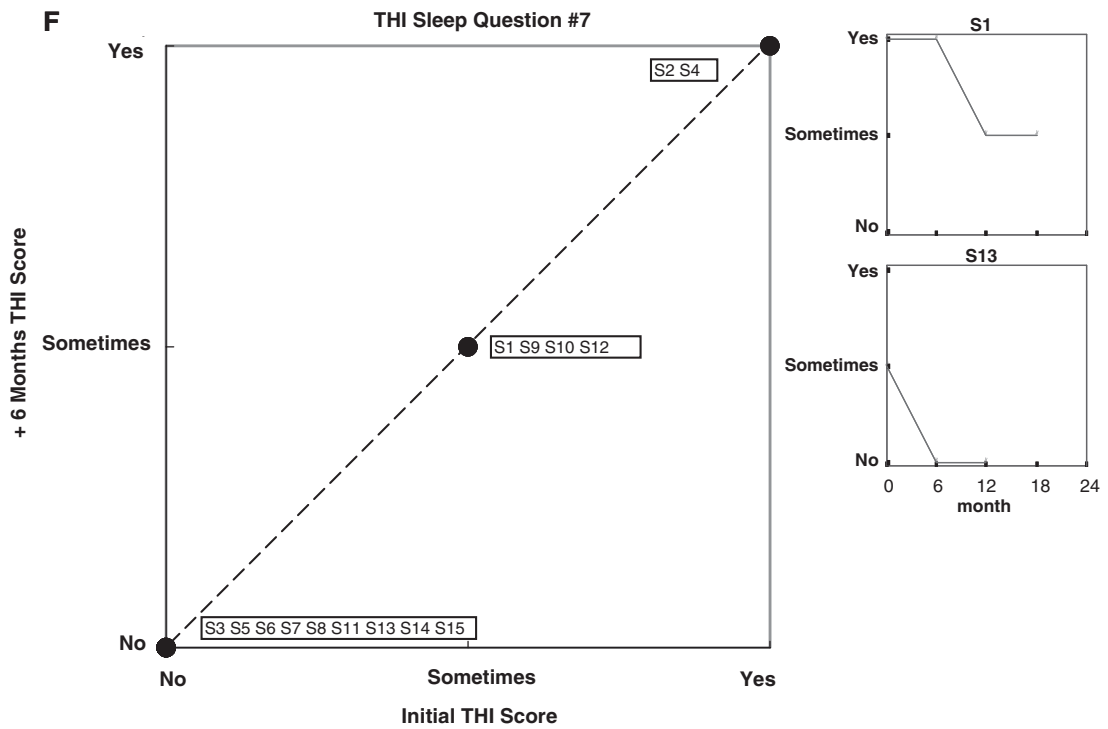


Fig. 3 (Continued)

This issue seems to arise in part from the analysis of group data. As the number of participants increases, the standard error decreases, and the test score required to reach statistical significance becomes smaller. People ask whether these small significant differences are really clinically relevant.

We believe this is the wrong question, and arises from confusion about individual versus group effects. In the example above, the statistical test is evaluating the likelihood that the scores arose from two different groups. A statistically significant difference between groups suggests that the groups are different. Of course, what is considered "significant" often depends on which statistical methods are used.

A different approach to this issue is to consider the validity of the questionnaire. If the questionnaire is designed to measure clinically real issues and it is valid, then we believe that a statistically significant difference for an individual is (by definition) clinically meaningful.

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